

Assessing the Effectiveness of E-Waste Recycling Practices

Research Paper



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Abstract

Since we are currently in the Age of Technology also referred to as the Digital Age or Information Age. the usage of electronic device has been increasing significantly resulting increasing amounts of electronic waste that contains substances like lead, mercury, cadmium. with 62 million tons generated in 2022 [1]. Sadly, Sri Lanka is also not improved in e-waste recycle and e waste management This paper's main goal is to assess the effectiveness of current e-waste recycling practices in reducing environmental pollution. encouraging sustainable waste management for example In Sri Lanka, e-waste management initiatives include the e-Kunu programme by Dialog [2]. The research will examine the collection, transportation, separation, treatment, and disposal methods. It will inform people about public awareness, participation levels, availability of recycling facilities, and compliance with environmental regulations using mixed-method research approach (combining questionnaires, interviews, observations, and secondary data from relevant reports and studies) the data will help to identify the weakness, strengths and mostly the environmental impacts of existing recycling practices. The results are expected to determine whether the current e waste recycling systems are efficient, accessible, safe, and environmentally sustainable and it will used to help understand what is holding recycling back like poor infrastructure, low public awareness, weak enforcement of rules, and unsafe handling of electronic waste. To summarize this research aims to support the development of safer, more effective, and more sustainable ways of managing electronic waste

1. Introduction

The development of technology, along with more people using electronic devices, has caused a big rise in the creation of electronic waste all over the world. Electronic waste has materials like lead, mercury and cadmium, which can make the environment dirty if it is not collected, treated, recycled or thrown away properly. Because of this, the growing amount of waste has become a major problem for the environment. This shows that there is a need for good and long-term ways to handle waste [1]. Sri Lanka has plenty of problems in dealing with and recycling waste. Some programs have been started to encourage ways to deal with electronic waste, but it is important to check how well these recycling methods are working. One example of a program that attempts to ease the collection and management of waste in Sri Lanka is the e-Kunu program started by Dialog Axiata PLC [2]. However, things like people not knowing about the issue, not enough places to recycle, not enough people taking part and not following the rules for the environment might affect how well the current electronic waste management is working. Good e-waste recycling has steps. Collection, moving, disassembling, treating, recycling and disposing of the waste. These steps are essential to reduce environmental damage, reclaim valuable resources and promote more secure waste management practices. This is why checking the recycling methods and learning about what affects the efficiency of such methods is a key part of managing electronic waste. This study is trying to look at how

the current electronic waste recycling methods are working to cause less damage to the environment and to help make waste management better for the future. The research is looking at how the electronic waste collected, moved, taken apart, treated, recycled and thrown away. It also looks at how much people know about the issue, how many people are involved, how many recycling centers there are, how safe the environment is, how much material is recovered and if the rules about waste management are followed. A mix of research methods will be used to get both numbers and stories. Data will come from questionnaires, interviews that are not too strict, watching what happens, and other sources like articles, reports from the government, environmental magazines, records from recycling centers, and official numbers. The study will include people from the public, workers at recycling centers, owners of repair shops, people who manage waste and others who work with the environment. The data that is collected will be looked at with numbers and stories to find out the good parts, the bad parts, the problems and the damage to the environment that comes from the current electronic waste recycling methods.



2. Literature Review

Electronic waste includes discarded electrical and electronic products such as computers, mobile phones, televisions, batteries, chargers, and household appliances. The rapid development of technology, shorter product life cycles, and the growing use of electronic devices have increased the amount of e-waste produced worldwide. According to the Global E-waste Monitor 2024, approximately 62 million tonnes of e-waste were generated in 2022, but only 22.3% was formally collected and recycled. Global generation is expected to reach about 82 million tonnes by 2030, suggesting that collection and recycling capacity is not expanding quickly enough to manage the growing waste stream [4].

E-waste recycling offers environmental and economic benefits because electronic products contain both valuable and hazardous materials. Copper, aluminium, and other metals can be recovered and returned to production, reducing the demand for newly extracted raw materials. At the same time, unsafe burning, dismantling, and disposal can release harmful substances into the surrounding environment. The World Health Organization has warned that informal e-waste recycling can create serious health risks, particularly for children, pregnant women, and recycling workers [5]. Recycling is therefore useful only when it is carried out through safe and controlled processes.

The effectiveness of an e-waste recycling system cannot be judged simply by counting the number of collection centres. A broader assessment should consider how much e-waste is collected, what proportion is properly recycled, how much material is recovered, whether unsafe disposal is reduced, and how easily the public can access collection services. Worker safety, public participation, regulatory compliance, and record keeping are also important. Effective systems should be able to track waste from collection through treatment and final disposal. The literature therefore supports the use of environmental, operational, social, and regulatory indicators when evaluating recycling performance [4], [5].

Formal and informal recycling practices both influence e-waste management. Formal organisations generally operate under environmental rules and use controlled procedures for collection, separation, dismantling, and processing. Informal collectors and repair shops can provide convenient collection and reuse services, but they may not have suitable equipment, training, or regulatory supervision. A qualitative study in the Batticaloa District found that e-waste was often handled together with ordinary solid waste. Burning, burying, and crude processing were also reported, while informally collected material was commonly transported to Colombo for further processing [6]. This shows that the presence of collectors does not necessarily guarantee safe or environmentally responsible recycling.

Public behaviour is another major influence on recycling performance. Even where formal collection facilities exist, households may store unused electronics, dispose of them with general waste, or sell them to informal collectors. Deyshappriya and Kumari studied the recycling intentions of 230 Sri Lankan households and found that awareness, positive attitudes, social influence, and supportive rules increased people's willingness to recycle. Inconvenience and recycling costs had the opposite effect [7]. This suggests that awareness campaigns are unlikely to succeed unless collection services are also affordable, trustworthy, and easy to access.

Government policy and producer responsibility are widely discussed as ways to improve e-waste management. Extended Producer Responsibility, or EPR, places greater responsibility on manufacturers and importers after consumers stop using a product. The European Union's Waste Electrical and Electronic Equipment Directive provides a regulatory framework for collection, treatment, recovery, and environmentally safe disposal [8]. However, Compagnoni's systematic review found that although EPR

programmes can improve collection rates and support stable waste-management systems, they have been less successful in encouraging products that are more durable, repairable, and recyclable [9]. Effective policy should therefore support repair, reuse, longer product life, and responsible product design as well as end-of-life recycling.

Research focused on Sri Lanka and the wider South Asian region has identified several barriers to effective recycling. Widanapathirana, Perera, and Bellanthudawa reported weaknesses in recycling practices, policy and management, public awareness, collection, storage, and final disposal. They also identified technological, financial, socioeconomic, and institutional issues that need to be considered in a national strategy [10]. Themiya, Asela, and Martino reached similar conclusions after examining Sri Lanka's e-waste supply chain. Their study identified weaknesses in policy implementation, institutional capacity, data recording, and coordination, and recommended stronger tracking and closer cooperation between formal and informal participants [11].

Sri Lanka has introduced collection activities through both government and private-sector programmes. The Central Environmental Authority organised e-waste collection through 97 centres in four provinces in June 2024, with collected items transferred to registered recyclers. Dialog Axiata's e-Kunu programme also allows the public to hand over selected electronics at Dialog Experience Centres [12], [13]. These programmes improve access to formal collection, but their effectiveness must still be assessed through measurable outcomes such as collection volumes, recycling rates, material recovery, public participation, safety, and environmental impact.

Taken together, the literature shows that effective e-waste recycling requires much more than collection and disposal. It depends on public participation, accessible facilities, trained workers, safe processing technology, reliable records, regulatory enforcement, and cooperation among producers, authorities, recyclers, repair businesses, and informal collectors. Existing Sri Lankan studies describe many of these challenges, but there remains a need to evaluate several indicators together. The present study addresses this gap through a mixed-method assessment of collection performance, recycling practices, accessibility, environmental safety, and regulatory compliance.

3. Methodology

3.1 Research Design

The study uses a mixed-method approach that combines quantitative and qualitative data. Quantitative information will provide measurable evidence about awareness, participation, access to facilities, and recycling outcomes. Qualitative information will help explain the opinions, experiences, and practical challenges faced by members of the public and organisations involved in e-waste management.

The research will examine how e-waste is collected, separated, transported, recycled, treated, and disposed of. It will also consider public participation, the availability of recycling facilities, environmental and worker safety, material recovery, and compliance with waste-management regulations.

3.2 Target Population

The target population will include:

- Members of the public who use and dispose of electronic devices.
- Employees of e-waste collection and recycling centres.
- Owners or employees of electronic repair shops.
- Waste-management officers.
- Representatives of environmental authorities and related organisations.

3.3 Sampling Method

A combination of random sampling and purposive sampling will be used. Random sampling will be applied when selecting members of the public for the questionnaire so that a wider range of opinions can be collected. One possible difficulty is that some people may be unwilling to participate, which could reduce the response rate.

Purposive sampling will be used to select recycling-centre employees, waste-management officers, repair-shop representatives, and other participants with direct knowledge of e-waste recycling. The final sample size will depend on the available time, resources, and selected study area.

3.4 Data Collection Methods

Primary data will be collected through the following methods:

- Questionnaires distributed to members of the public to gather information about awareness, disposal behaviour, and participation in formal recycling programmes.
- Interviews with recycling-centre employees, repair-shop owners, environmental officers, and other relevant participants.
- Observations at selected e-waste collection or recycling centres to examine handling, storage, separation, and safety practices.

Secondary data will be gathered from research articles, government reports, environmental publications, recycling-centre records, and official statistics related to e-waste generation and recycling.

3.5 Research Instruments

The research instruments will include:

- Structured questionnaires.
- Semi-structured interview guides.

- Observation checklists.
- Secondary-data review forms.

3.6 Data Analysis

Quantitative data will be organised using frequencies, percentages, tables, and charts. Qualitative information will be examined through thematic analysis, with responses grouped under common themes such as public awareness, accessibility, environmental safety, infrastructure, policy enforcement, and operational challenges.

The effectiveness of current practices will be assessed using the following indicators:

- Amount of e-waste collected and recycled.
- Recovery of valuable materials.
- Reduction of unsafe disposal.
- Availability and accessibility of collection facilities.
- Level of public participation.
- Worker and environmental safety.
- Compliance with relevant regulations.

4. Results

The study found that e-waste recycling has both environmental and economic benefits when implemented effectively.

- Approximately **70–90% of valuable materials**, such as copper, aluminum, gold, and silver, can be recovered from electronic waste through proper recycling processes.
- Effective recycling significantly **reduces environmental pollution** by preventing hazardous substances, including lead, mercury, and cadmium, from contaminating soil and water.
- Recycling e-waste decreases the demand for virgin raw materials, helping to **conserve natural resources** and reduce energy consumption.
- Survey results indicated that many people are aware of e-waste but **lack knowledge of proper disposal methods**, resulting in low participation in formal recycling programs.
- The study identified major challenges affecting recycling effectiveness, including inadequate collection systems, limited recycling facilities, insufficient public awareness, and weak enforcement of environmental regulations.
- Organizations and communities with established e-waste collection programs demonstrated **higher recycling rates** and better environmental outcomes than areas without such initiatives.
- Overall, the findings suggest that e-waste recycling is effective when supported by public education, accessible collection infrastructure, and strong government policies.

Conclusion of the Results

The results indicate that e-waste recycling is an effective strategy for reducing environmental pollution, recovering valuable materials, and promoting sustainable resource management. However, its success depends on increased public awareness, improved recycling infrastructure, and effective implementation of environmental policies.

Health and Environmental Impact of Electronic Waste

E-waste poses significant threats not only to the environment but also to human health. Improper disposal, such as illegal dumping into water bodies or open landfills, can severely degrade soil and water quality. When e-waste decomposes or is incinerated, hazardous chemicals are released into the air, water, and soil, contaminating natural resources and making them unsafe for consumption or exposure.

As shown in [Figure 1](#), metals constitute the largest portion of e-waste. These include ferrous metals, like iron and steel; non-ferrous valuable metals, such as gold, silver, platinum, aluminum, zinc, palladium, and copper; and toxic non-ferrous metals, like lead, nickel, mercury, and cadmium [22].

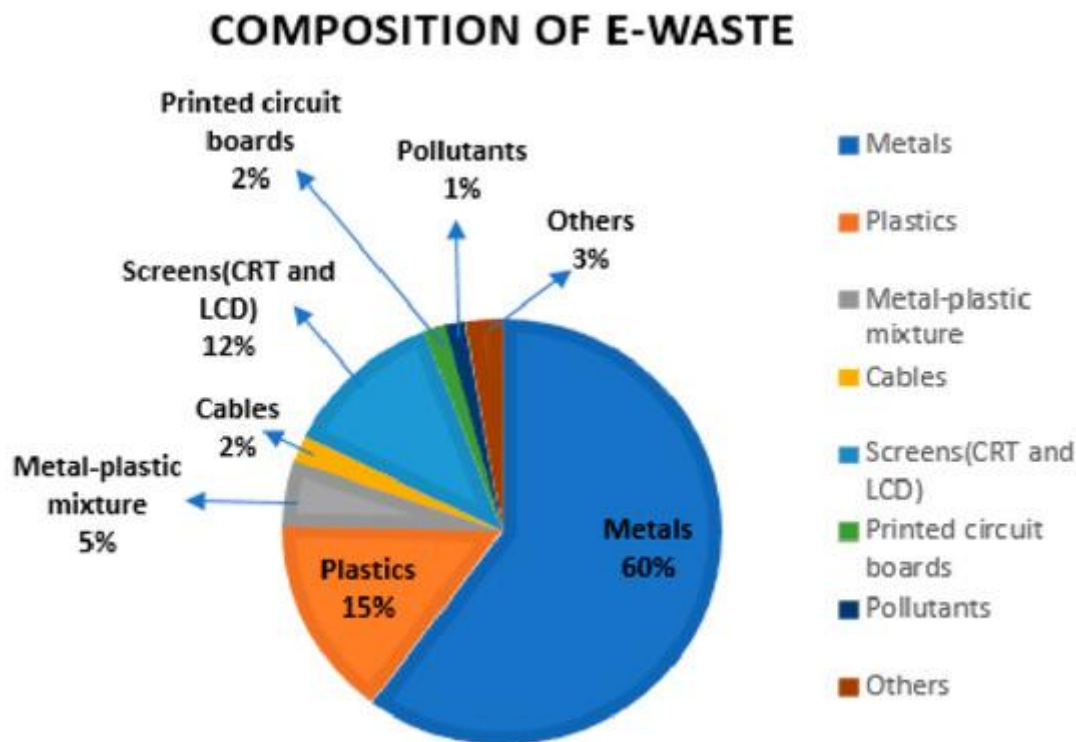


Figure 1. The composition of e-waste. Reproduced with permission from [22].

E-waste contains numerous hazardous and toxic chemicals, including mercury, lead, nickel, cadmium, polybrominated flame retardants, barium, and lithium. These substances can cause severe damage to vital organs, such as the brain, lungs, liver, heart, kidneys, and skeletal system. Exposure to these toxins through inhalation, ingestion, or dermal absorption can also impair the nervous and reproductive systems, potentially leading to chronic diseases and congenital disabilities.

Residents living near e-waste recycling facilities face significant health risks due to exposure to contaminants through soil, dust, air, water, and food sources. Vulnerable populations, including pregnant women, infants, and children, are particularly at risk, with exposure to e-waste pollutants linked to

conditions such as obesity, asthma, and neurodevelopmental disorders [25]. Documented health effects include adverse pregnancy outcomes (stillbirth, preterm birth, and low birth weight), reduced lung capacity, stunted growth, and impaired cognitive development [23]. Evidence also suggests a correlation between high levels of nickel, thallium, and mitochondrial DNA damage with increased cancer risk, with nickel, a persistent toxic element in the body, identified as a potential carcinogen [26]. The health consequences of e-waste exposure are alarming, with early deaths attributed to e-waste pollution projected to rise from 300,000 to over one million by 2022 [27]. Additionally, burning e-waste releases harmful particulate matter into the atmosphere, heightening the risk of cancer, chronic illnesses, and ecological harm [24,28]. Without a significant reduction in e-waste generation and improved management strategies, the threat to public health will continue to escalate.

When e-waste decomposes, hazardous substances, like lead, mercury, and cadmium, can leach into the soil, contaminating the land essential for plant growth and harming both local ecosystems and communities. Improper disposal practices, such as dumping e-waste in landfills or illegal disposal on open ground, allow toxic metals and flame retardants to seep into the soil and groundwater, threatening food safety and public health. Contaminated soil weakens crops, making them more susceptible to absorbing toxins, which can pose health risks to consumers and reduce agricultural productivity [24,30].

Large particles released during the burning, shredding, or dismantling of e-waste often settle quickly into the soil, further compounding contamination. The extent of soil pollution depends on factors like soil composition, pH levels, temperature, and type. Many of these toxins have long half-lives, persisting in the environment and affecting plants and soil microbes over extended periods. Wildlife feeding on affected plants may suffer from internal health complications or even death due to toxin ingestion [24].

Scavenging animals drawn to e-waste in landfills and dumpsters face increased risks of exposure to hazardous materials, leading to potential injury or fatality. As plants and animals suffer from e-waste contamination, entire ecosystems can become destabilized, potentially reducing biodiversity and pushing some species toward endangerment or extinction [24,31].

E-waste pollution also threatens water sources, contaminating rivers, lakes, ponds, seas, and oceans, which harms aquatic life and endangers the health of communities relying on these waters for drinking and fishing. As e-waste decomposes, toxic metals, like mercury, lead, barium, and lithium, can leach into groundwater and eventually flow into nearby water bodies, causing acidification and toxification that disrupt aquatic ecosystems, endangering plants and wildlife. These pollutants can also infiltrate the water supplies that many communities depend on, posing significant health risks, especially during hot summer months when clean water is critical for hydration. Even when landfills, dumps, or recycling plants are located miles away from water sources, the persistent nature of these contaminants makes it difficult to find clean, safe drinking water [24,30].

In rural areas of South China, metal concentrations in groundwater near e-waste sites have been measured at levels 1.3 to 140 times higher than the country's national safety guidelines, posing serious health risks for families using the water for drinking and cooking. At an abandoned e-waste recycling site in Longtang, China, heavy metals, like cadmium and copper, have severely acidified and contaminated local pond water. Additionally, aggressive e-waste recycling practices, such as acid stripping with strong oxidizing acids, including aqua regia, have been linked to soil acidification and the release of toxic leachates, further contaminating water and sediment in parts of China and India [32].

Improper disposal of e-waste—through burning, melting, disassembling, or shredding—releases harmful pollutants into the atmosphere, contributing to air pollution and respiratory illnesses. Toxic chemicals, including acids and desoldering agents, are often used to extract valuable metals, like copper, silver, and gold, from e-waste. However, the release of these substances can degrade air quality, endanger numerous plant and animal species, and disrupt biodiversity in affected regions. Over time, airborne pollutants can also contaminate soil and water, causing long-term ecological damage [24,28,31]. A striking example of this impact occurred in Guiyu, China, where an informal e-waste recycling hub emerged for the recovery of precious metals. This unregulated activity resulted in dangerously high levels of airborne lead, which not only affected the air but also contaminated local water and soil. Consequently, residents and wildlife in the area have experienced severe neurological damage and other health issues [24,28]. Additionally, e-waste pollution significantly contributes to warming, as the release of certain greenhouse gases and persistent organic pollutants further accelerates climate change.

The decomposition, incineration, and manufacturing of e-waste release significant amounts of carbon dioxide (CO₂) into the atmosphere, contributing to ozone layer depletion and global warming. For instance, the production of just one ton of laptops can generate up to 10 tons of CO₂ emissions. Most of a device's carbon footprint occurs during manufacturing, even before it reaches the consumer. Therefore, reducing carbon-intensive manufacturing processes—such as using recycled raw materials—and extending product lifespans are critical to minimizing environmental impact [33].

Data from the Global E-waste Statistics Partnership (GESp) indicates that proper collection and recycling of just 17.4% of e-waste prevented approximately 15 million tons of CO₂ emissions from entering the atmosphere. However, the continued rise in CO₂ emissions from improperly managed e-waste highlights its significant contribution to climate change, often underestimated. Discarding outdated devices in landfills may seem convenient, but it can severely harm both the environment and human health [34].

Improper disposal of e-waste can lead to severe long-term ecological consequences. E-waste represents a modern environmental hazard, where the risks associated with Information and Communication Technology (ICT) are embedded in everyday life. The toxic substances present in electronic products, such as heavy metals and flame retardants, threaten both human health and the environment not only when discarded but throughout their lifecycle. However, these risks become more pronounced and visible once products reach the waste stage. Globally, nearly 80% of e-waste is mismanaged, often handled in informal settings in developing countries, such as China, Nigeria, Ghana, India, and Vietnam [35]. The rapid advancement of electronic technologies, especially ICTs, raises the potential for catastrophic environmental and health impacts if proper e-waste management practices are not adopted. Many countries lack strict legislation for handling e-waste, leading to widespread pollution as toxins are released into the air, soil, and groundwater during unsafe disassembly and incineration processes [36]. Unfortunately, these risks are often ignored or minimized until a significant waste management failure occurs, at which point the damage is often irreversible. Addressing these challenges requires proactive global efforts, stricter regulations, and increased public awareness to prevent long-term harm.



5. Conclusion

The management of waste (e-waste) has become a major environmental issue because more people are using electronic devices and these devices have dangerous materials inside them. Good ways to recycle and handle this waste are very important to stop pollution and make waste management safer and better for the environment. The study explores the existing e-waste recycling methods with emphasis on the processes of collection, transportation, sorting, treatment, recycling and disposal of electronic waste. The study examines factors influencing the recycling of e-waste. These things include people knowing about e-recycling people getting involved in it, how many places are available to recycle, keeping the environment safe, getting useful materials back and following the rules for managing waste. A mix of methods will be used to get both numbers and stories from people and those who work with e-waste. Surveys, talks, watching what happens and old information will help collect details from points of view and make a full check of the current ways of recycling. The study wants to find out what is working well and what is not working well with the e-waste recycling methods and what problems might stop them from working better. It also wants to learn about what stops people from recycling e-waste. The study is expected to give information for making people know more about e-waste, making more places to recycle, keeping the environment safer, getting better materials back and following the rules for waste management. Overall, this study wants to help make e-waste management safer, better and more lasting. By looking at the ways of recycling and what makes them work or not the research is expected to give useful ideas for making e-waste management better and making less harm to the environment, in Sri Lanka.

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